#	NODE	TERMINAL (T) / NO TERMINAL (NT)	TYPE (DUPLICATION / DELETION / MODIFICATION / VALID)	ID TEST	DESCRIPTION	FILE PATH	FILE CONTENT	EXPECTED RESULT	OBSERVATIONS
1	1-39,42	T/NT	VALID	TC1	Cover as many valid nodes as possible	resources/register_document/valid/tc1-valid_pdf.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	1a11adad20dead4345f337a775f3927f78432c1f6441608bb953a2df5133ec41	Must further test terminal nodes from <EXTENSION>
2	40	T	VALID	TC2	Testing .docx branch	resources/register_document/valid/tc2-valid_docx.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.docx" }	caf9f4c756850f8196a4976179a38d6c8d7ea8db3da0bca74b1d0e1e7d8352f0	Must test one more valid <EXTENSION> case
3	41	T	VALID	TC3	Testing .xlsx branch	resources/register_document/valid/tc2-valid_xlsx.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.xlsx" }	85392121ec2f93860df8697e9bd6bb7cb9e27cb458b8b3b8964b230139182f66	All valid cases have been tested
4	1	NT	DELETION	TC4	Invalid JSON format from empty file	resources/register_document/invalid/tc4-empty_file.json		EnterpriseManagementException; The file is not JSON formatted.	Deleting node 2 and 4 would produce the same result
5	2	NT	DELETION	TC5	Invalid JSON format from missing curly bracket at the start	resources/register_document/invalid/tc5-missing_json_start.json	"PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	Deleting node 1 and 4 would produce the same result
6	3	NT	DELETION	TC6	Invalid JSON from missing FIELDS	resources/register_document/invalid/tc6-missing_fields.json	{ }	EnterpriseManagementException; JSON does not have the expected structure: missing <FIELDS>	Deleting the <FIELDS> node will still keep the JSON start and end brackets
7	4	NT	DELETION	TC7	Invalid JSON format from missing curly bracket at the end	resources/register_document/invalid/tc7-missing_json_end.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf"	EnterpriseManagementException; The file is not JSON formatted.	Deleting node 1 and 2 would produce the same result
8	6	NT	DELETION	TC8	Invalid JSON from missing PROJECT_ID	resources/register_document/invalid/tc8-missing_project_id.json	{ , "FILENAME": "File1234.xlsx" }	EnterpriseManagementException; The file is not JSON formatted.	Deleting node 6 has the same effect as nodes 10-16, and 26-32
9	7	NT	DELETION	TC9	Invalid JSON from missing SEPARATOR	resources/register_document/invalid/tc9-missing_separator.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2" "FILENAME": "File1234.docx" }	EnterpriseManagementException; The file is not JSON formatted.	Deleting node 7 produces the same effect as duplicating/modifying node 17 as they all cause an invalid JSON format
10	8	NT	DELETION	TC10	Invalid JSON from missing FILENAME	resources/register_document/invalid/tc10-missing_filename.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", }	EnterpriseManagementException; The file is not JSON formatted.	Deleting node 8 produces the same effect as deleting node 18-25 and 33-42
11	6	NT	DUPLICATION	TC11	Invalid JSON from duplicate <PROJECT_ID>	resources/register_document/invalid/tc8-duplicate_project_id.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "PROJECT_ID": "f5155610659bc5c9e5cc9a4e4ed8dd5f", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; JSON does not have the expected structure: duplicate field <PROJECT_ID>	Yields the same effect as duplicating the FILENAME field (node 8) or SEPARATOR field value (node 17)

12	8	NT	DUPLICATION	TC12	Invalid JSON from duplicate <FILENAME>	resources/register_document/invalid/tc9-duplicate_filename.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf", "FILENAME": "File5678.pdf" }	EnterpriseManagementException; JSON does not have the expected structure: duplicate field <FILENAME>	Yields the same effect as duplicating the PROJECT_ID field (node 6) or SEPARATOR field value (node 17)
13	10	NT	DELETION	TC13	Invalid JSON format from missing opening quotations for PROJECT_ID label	resources/register_document/invalid/tc13_del_open_quote_proj_id_label.json	{ PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	Same effect as deleting all other quotations nodes (12,14,16.18,20,22,25); specifically yields the same effect as deleting FILENAME label opening quotations (node 18)
14	11	NT	DELETION	TC14	Invalid JSON format from missing project <id_label>	resources/register_document/invalid/tc14_missing_proj_id_label.json	{ "": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; JSON does not have the expected structure. Invalid PROJECT_ID label	Same exception handling as duplicating/modifying node 27; yields same effect as deleting FILENAME field label (node 19)
15	12	NT	DELETION	TC15	Invalid JSON format from missing ending quotations for PROJECT_ID label	resources/register_document/invalid/tc13_del_close_quote_proj_id_label.json	{ "PROJECT_ID: "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	Same effect as deleting FILENAME label ending quotations (node 20)
16	13	NT	DELETION	TC16	Invalid JSON format from missing colon within PROJECT_ID field	resources/register_document/invalid/tc16_missing_colon_project_id.json	{ "PROJECT_ID" "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	Same effect as deleting FILENAME field's colon (node 21)
17	14	NT	DELETION	TC17	Invalid JSON format from missing opening quotations for PROJECT_ID value	resources/register_document/invalid/tc17_del_open_quote_proj_id_value.json	{ "PROJECT_ID": deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	Same effect as deleting FILENAME's value opening quotations (node 22)
18	15	NT	DELETION	TC18	Missing project <id_value> causes "JSON data has no valid values" error	resources/register_document/invalid/tc18_missing_proj_id_value.json	{ "PROJECT_ID": "", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; JSON data has no valid values: invalid PROJECT_ID value	Yields same exception message as duplicating/modifying node 31
19	16	NT	DELETION	TC19	Invalid JSON format from missing ending quotations for PROJECT_ID value	resources/register_document/invalid/tc19_del_close_quote_proj_id_value.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2, "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	Same effect as deleting FILENAME's value closing quotations (node 25)
20	18	NT	DELETION	TC20	Invalid JSON format from missing opening quotations for FILENAME label	resources/register_document/invalid/tc20_del_open_quote_filename_label.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	Yields the same effect as deleting PROJECT_ID label opening quotations (node 10)
21	19	NT	DELETION	TC21	Invalid JSON format from missing <filename_label>	resources/register_document/invalid/tc21_missing_filename_label.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "": "File1234.pdf" }	EnterpriseManagementException; JSON does not have the expected structure. Invalid FILENAME label	Same exception handling as duplicating/modifying node 34; yields same effect as deleting PROJECT_ID field label (node 11)
22	20	NT	DELETION	TC22	Invalid JSON format from missing ending quotations for FILENAME label	resources/register_document/invalid/tc22_del_close_quote_filename_label.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME: "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	Same effect as deleting PROJECT_ID label ending quotations (node 12)

23	21	NT	DELETION	TC23	Invalid JSON from missing colon for FILENAME field	resources/register_document/invalid/tc23_missing_colon_filename.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME" "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	Same effect as deleting PROJECT_ID field's colon (node 13)
24	22	NT	DELETION	TC24	Invalid JSON format from missing opening quotations for FILENAME value	resources/register_document/invalid/tc24_del_open_quote_filename_value.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	Same effect as deleting PROJECT_ID's value opening quotations (node 14)
25	23	NT	DELETION	TC25	Invalid JSON format from missing NAME	resources/register_document/invalid/tc25_missing_name.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": ".pdf" }	EnterpriseManagementException; JSON data has no valid values: invalid NAME	This yields the same exception message as duplicating/modifying node 38
26	24	NT	DELETION	TC26	Invalid JSON format from missing EXTENSION	resources/register_document/invalid/tc25_missing_extension.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234" }	EnterpriseManagementException; JSON data has no valid values: invalid EXTENSION	Yields same result as duplicating extension terminal nodes (39, 40, 41)
27	25	NT	DELETION	TC27	Invalid JSON format from missing ending quotations for FILENAME value	resources/register_document/invalid/tc26_del_close_quote_filename_value.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf }	EnterpriseManagementException; The file is not JSON formatted.	Same effect as deleting PROJECT_ID's value closing quotations (node 16)
28	5	T	DUPLICATION	TC28	Duplicate JSON start bracket causes JSON format error	resources/register_document/invalid/tc28_dup_json_start.json	{{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This produces the same result as duplicating node 9
29	5	T	MODIFICATION	TC29	JSON format error as a curly bracket is expected to start a JSON file, not a square bracket	resources/register_document/invalid/tc29_mod_json_start.json	["PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This produces the same result as modifying node 9
30	9	T	DUPLICATION	TC30	Duplicate JSON end bracket causes JSON format error	resources/register_document/invalid/tc30_dup_json_end.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }}	EnterpriseManagementException; The file is not JSON formatted.	This produces the same result as duplicating node 5
31	9	T	MODIFICATION	TC31	JSON format error as a curly bracket is expected to end a JSON file, not a square bracket	resources/register_document/invalid/tc31_mod_json_end.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf"]	EnterpriseManagementException; The file is not JSON formatted.	This produces the same result as modifying node 5
32	26	T	DUPLICATION	TC32	Duplicate opening quotation mark for PROJECT_ID label causes JSON format error	resources/register_document/invalid/tc32_dup_open_quote_proj_id_label.json	{ ""PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as duplicating the opening quotations node in the filename field key's label (node 33). Produces same exception error as the duplication of all other quotations nodes (28, 30, 32, 33, 35, 37, 42)
33	26	T	MODIFICATION	TC33	JSON format error as a double quotation mark is expected before PROJECT_ID label	resources/register_document/invalid/tc33_mod_open_quote_proj_id_label.json	{ 'PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as modifying the opening quotations node in the filename field key's label (node 33). Produces same exception error as the modification of all other quotations nodes (28, 30, 32, 33, 35, 37, 42)

34	27	T	DUPLICATION	TC34	Duplicating node 27 yields an invalid label	resources/register_document/invalid/shared-invalid_project_id_label.json	{ "PROJECT_IDPROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; JSON does not have the expected structure. Invalid PROJECT_ID label	This yields the same effect as duplicating/modifying node 34, and has same exception handling as modifying node 27 and deleting node 11 (all yield an invalid label for PROJECT_ID)
35	27	T	MODIFICATION	TC35	Modifying the field name for project ID yields an invalid label	resources/register_document/invalid/shared-invalid_project_id_label.json	{ "PROJECT_IDPROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; JSON does not have the expected structure. Invalid PROJECT_ID label	produces the same effect as modifying/duplicating the FILENAME field label (node 34), and duplicating node 27
36	28	T	DUPLICATION	TC36	Duplicate ending quotation mark for PROJECT_ID label causes JSON format error	resources/register_document/invalid/tc36_dup_close_quote_project_id_label.json	{ "PROJECT_ID"": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as duplicating the closing quotations node in the filename field key's label (node 35). Produces same exception error as the duplication of all other quotations nodes (26, 30, 32, 33, 35, 37, 42)
37	28	T	MODIFICATION	TC37	JSON format error as a double quotation mark is expected after PROJECT_ID label	resources/register_document/invalid/tc37_mod_close_quote_project_id_label.json	{ "PROJECT_ID': "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as modifying the closing quotations node in the filename field key's label (node 35). Produces same exception error as the modification of all other quotations nodes (26, 30, 32, 33, 35, 37, 42)
38	29	T	DUPLICATION	TC38	Duplicating the colon in the PROJECT_ID field violates JSON format	resources/register_document/invalid/tc38_dup_colon_project_id.json	{ "PROJECT_ID":: "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as duplicating node 36
39	29	T	MODIFICATION	TC39	JSON format error as a colon is expected, not an equal sign	resources/register_document/invalid/tc39_mod_colon_project_id.json	{ "PROJECT_ID"= "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as modifying node 36
40	30	T	DUPLICATION	TC40	Duplicate opening quotation mark for PROJECT_ID value causes JSON format error	resources/register_document/invalid/tc40_dup_open_quote_project_id_value.json	{ "PROJECT_ID": ""deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as duplicating the opening quotations node in the filename field's value (node 37). Produces same exception error as the duplication of all other quotations nodes (26, 28, 32, 33, 35, 37, 42)
41	30	T	MODIFICATION	TC41	JSON format error as a double quotation mark is expected before PROJECT_ID value	resources/register_document/invalid/tc41_mod_open_quote_project_id_value.json	{ "PROJECT_ID": 'deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as modifying the opening quotations node in the filename field's value (node 37). Produces same exception error as the modification of all other quotations nodes (26, 28, 32, 33, 35, 37, 42)
42	31	T	DUPLICATION	TC42	Duplication causes the value for PROJECT_ID to not follow MD5 standards, thus yielding "JSON data has no valid values" error	resources/register_document/invalid/shared-invalid_project_id_value.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; JSON data has no valid values: invalid PROJECT_ID value	This yields the same effect as duplicating/modifying nodes 38,39,40,41; and gets same exception message as modifying node 31 or deleting node 15

43	31	T	MODIFICATION	TC43	Modification causes the value for PROJECT_ID to not follow MD5 standards, thus yielding "JSON data has no valid values" error	resources/register_document/invalid/shared-invalid_project_id_value.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; JSON data has no valid values: invalid PROJECT_ID value	This yields the same result as duplicating node 31, so we can reference the same JSON path in our test
44	32	T	DUPLICATION	TC44	Duplicate ending quotation mark for PROJECT_ID value causes JSON format error	resources/register_document/invalid/tc44_dup_close_quote_project_id_value.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2"", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as duplicating the closing quotations node in the filename field's value (node 42). Produces same exception error as the duplication of all other quotations nodes (26, 28, 30, 33, 35, 37, 42)
45	32	T	MODIFICATION	TC45	JSON format error as a double quotation mark is expected after PROJECT_ID value	resources/register_document/invalid/tc45_mod_close_quote_project_id_value.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2', "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as modifying the closing quotations node in the filename field's value (node 42). Produces same exception error as the duplication of all other quotations nodes (26, 28, 30, 33, 35, 37, 42)
46	17	T	DUPLICATION	TC46	Invalid JSON format as a single comma is expected to separate fields	resources/register_document/invalid/tc46_dup_separator.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2",, "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	Produces the same result as modifying or deleting node 17 as they all cause an invalid JSON format
47	17	T	MODIFICATION	TC47	Invalid JSON format as a single comma is expected to separate fields	resources/register_document/invalid/tc47_mod_separator.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2"; "FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	Produces the same result as duplicating or deleting node 17 as they all cause an invalid JSON format
48	33	T	DUPLICATION	TC48	Duplicate opening quotation mark for FILENAME label causes JSON format error	resources/register_document/invalid/tc48_dup_open_quote_filename_label.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", ""FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as duplicating the opening quotations node in the project id field's label (node 26). Produces same exception error as the duplication of all other quotations nodes (26, 28, 30, 32, 35, 37, 42)
49	33	T	MODIFICATION	TC49	JSON format error as a double quotation mark is expected before FILENAME label	resources/register_document/invalid/tc49_mod_open_quote_filename_label.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", 'FILENAME": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as modifying the opening quotations node in the project id field's label (node 26). Produces same exception error as the modification of all other quotations nodes (26, 28, 30, 32, 35, 37, 42)
50	34	T	DUPLICATION	TC50	Duplicating node 34 yields an invalid field label	resources/register_document/invalid/shared-invalid_filename_label.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAMEFILENAME": "File1234.pdf" }	EnterpriseManagementException; JSON does not have expected structure. Invalid FILENAME label	Duplication and modification of node 34 can both result in an identical malformed key, allowing us to use the same file path for optimizing efficiency. Both cause a JSON expected structure error
51	34	T	MODIFICATION	TC51	Modifying the field name for project ID yields an invalid label	resources/register_document/invalid/shared-invalid_filename_label.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAMEFILENAME": "File1234.pdf" }	EnterpriseManagementException; JSON does not have expected structure. Invalid FILENAME label	Duplication and modification of node 34 can both result in an identical malformed key, allowing us to use the same file path for optimizing efficiency. Both cause a JSON expected structure error

52	35	T	DUPLICATION	TC52	Duplicate ending quotation mark for FILENAME label causes JSON format error	resources/register_document/invalid/tc52_dup_close_quote_filename_label.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME"": "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as duplicating the closing quotations node in the project id field's label (node 28). Produces same exception error as the duplication of all other quotations nodes (26, 28, 30, 32, 33, 37, 42)
53	35	T	MODIFICATION	TC53	JSON format error as a double quotation mark is expected after FILENAME label	resources/register_document/invalid/tc53_mod_close_quote_filename_label.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME': "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as modifying the closing quotations node in the project id field's label (node 28). Produces same exception error as the modification of all other quotations nodes (26, 28, 30, 32, 33, 37, 42)
54	36	T	DUPLICATION	TC54	JSON format error from double colons after FILENAME	resources/register_document/invalid/tc54_dup_colon_filename.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME":: "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as duplicating node 29
55	36	T	MODIFICATION	TC55	JSON format error as a colon is expected after FILENAME label, not an equal sign	resources/register_document/invalid/tc55_mod_colon_filename.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME"= "File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as modifying node 29
56	37	T	DUPLICATION	TC56	Duplicate opening quotation mark for FILENAME value causes JSON format error	resources/register_document/invalid/tc56_dup_open_quote_filename_value.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": ""File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as duplicating the opening quotations node in the project id's value (node 30). Produces same exception error as the duplication of all other quotations nodes (26, 28, 30, 32, 33, 35, 42)
57	37	T	MODIFICATION	TC57	JSON format error as a double quotation mark is expected before FILENAME value	resources/register_document/invalid/tc57_mod_open_quote_filename_value.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": 'File1234.pdf" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as modifying the opening quotations node in the project id's value (node 30). Produces same exception error as the modification of all other quotations nodes (26, 28, 30, 32, 33, 35, 42)
58	38	T	DUPLICATION	TC58	Invalid JSON value from a modified file name that is not 8 alphanumeric chars	resources/register_document/invalid/shared-invalid_name.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234File1234.pdf" }	EnterpriseManagementException; JSON data has no valid values: invalid NAME	This yields the same effect as duplicating/modifying nodes 31; and gets same exception message as modifying node 38 or deleting node 23
59	38	T	MODIFICATION	TC59	Invalid JSON value from a modified file name that is not 8 alphanumeric chars	resources/register_document/invalid/shared-invalid_name.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234File1234.pdf" }	EnterpriseManagementException; JSON data has no valid values: invalid NAME	This yields the same result as duplicating node 31, so we can reference the same JSON path in our test
60	39	T	DUPLICATION	TC60	Having a duplicate extension violates the accepted extension types	resources/register_document/invalid/tc60_dup_pdf_extension.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf.pdf" }	EnterpriseManagementException; JSON data has no valid values: invalid EXTENSION	Yields same result as duplicating nodes 40 and 41
61	40	T	DUPLICATION	TC61	Having a duplicate extension violates the accepted extension types	resources/register_document/invalid/tc61_dup_docx_extension.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.docx.docx" }	EnterpriseManagementException; JSON data has no valid values: invalid EXTENSION	Yields same result as duplicating nodes 39 and 41

62	41	T	DUPLICATION	TC62	Having a duplicate extension violates the accepted extension types	resources/register_document/invalid/tc62_dup_xlsx_extension.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.xlsx.xlsx" }	EnterpriseManagementException; JSON data has no valid values: invalid EXTENSION	Yields same result as duplicating nodes 39 and 40
63	39	T	MODIFICATION	TC63	Invalid JSON value as .ppt is not a valid extension type for the FILENAME field	resources/register_document/invalid/shared-invalid_extension.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.ppt" }	EnterpriseManagementException; JSON data has no valid values: invalid EXTENSION	We can reuse this modified extension file for all modified extension test cases
64	40	T	MODIFICATION	TC64	Invalid JSON value as .ppt is not a valid extension type for the FILENAME field	resources/register_document/invalid/shared-invalid_extension.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.ppt" }	EnterpriseManagementException; JSON data has no valid values: invalid EXTENSION	We can reuse this modified extension file for all modified extension test cases
65	41	T	MODIFICATION	TC65	Invalid JSON value as .ppt is not a valid extension type for the FILENAME field	resources/register_document/invalid/shared-invalid_extension.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.ppt" }	EnterpriseManagementException; JSON data has no valid values: invalid EXTENSION	We can reuse this modified extension file for all modified extension test cases
66	42	T	DUPLICATION	TC66	Duplicate ending quotation mark for FILENAME value causes JSON format error	resources/register_document/invalid/tc66_dup_close_quote_filename_value.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf"" }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as duplicating the closing quotations node in the project id's value (node 32). Produces same exception error as the duplication of all other quotations nodes (26, 28, 30, 32, 33, 35, 37)
67	42	T	MODIFICATION	TC67	JSON format error as a double quotation mark is expected after FILENAME value	resources/register_document/invalid/tc67_mod_close_quote_filename_value.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf' }	EnterpriseManagementException; The file is not JSON formatted.	This yields the same effect as modifying the closing quotations node in the project id's value (node 32). Produces same exception error as the modification of all other quotations nodes (26, 28, 30, 32, 33, 35, 37)
68	N/A	N/A	N/A	TC68	Invalid from referencing a JSON path that does not exist	resources/register_document/invalid/tc68-file_does_not_exist.json	N/A	EnterpriseManagementException; Input file not found.	No file inside resources folder will be called "tc14-file_does_not_exist.json" in order to test catching this exception
69	N/A	N/A	N/A	TC69	Invalid case by forcing an internal processing error	resources/register_document/invalid/tc69-valid_for_signature_error.json	{ "PROJECT_ID": "deca2698d32baefe542ef5c3ba8236a2", "FILENAME": "File1234.pdf" }	EnterpriseManagementException; Internal processing error when getting the file_signature.	File Content is valid but inside TC16 we will mock failure while generating file_signature in order to throw this exception

#	PATH	ID TEST
1	1_3_H1(T)_H2(F)_H7_H1(T)_H2(F)_H7_H1(F)_H8_5(F)_7(F)_9(F)_11(F)_13(F)_15(F)_17(F)_19(F)_21(S)_23(S)_25_26_27_END	TC1
2	1(E)_2_END	TC68
3	1_3_4_END	TC4
4	1_3_H1(T)_H2(F)_H7_H1(T)_H2(T)_H3(T)_H4_END	TC11
5	1_3_H1(T)_H2(F)_H7_H1(T)_H2(T)_H3(F)_H5(T)_H6_END	TC12
6	1_3_H1(T)_H2(F)_H7_H1(T)_H2(T)_H3(F)_H5(F)_H7_H1(F)_H8_5(F)_7(T)_8_END	TC14
7	1_3_H1(F)_H8_5(T)_6_END	TC6
8	1_3_H1(T)_H2(F)_H7_H1(F)_H8_5(F)_7(T)_8_END	TC34
9	1_3_H1(T)_H2(F)_H7_H1(T)_H2(F)_H7_H1(F)_H8_5(F)_7(F)_9(T)_10_END	TC18
10	1_3_H1(T)_H2(F)_H7_H1(F)_H8_5(F)_7(F)_9(F)_11(T)_12_END	TC50
11	1_3_H1(T)_H2(F)_H7_H1(T)_H2(F)_H7_H1(F)_H8_5(F)_7(F)_9(F)_11(F)_13(T)_14_END	TC60
12	1_3_H1(T)_H2(F)_H7_H1(T)_H2(F)_H7_H1(F)_H8_5(F)_7(F)_9(F)_11(F)_13(F)_15(T)_16_END	TC65

13	1_3_H1(T)_H2(F)_H7_H1(T)_H2(F)_H7_H1(F)_H8_5(F)_7(F)_9(F)_11(F)_13(F)_15(F)_17(T)_18_END	TC61
14	1_3_H1(T)_H2(F)_H7_H1(T)_H2(F)_H7_H1(F)_H8_5(F)_7(F)_9(F)_11(F)_13(F)_15(F)_17(F)_19(T)_20_END	TC58
15	1_3_H1(T)_H2(F)_H7_H1(T)_H2(F)_H7_H1(F)_H8_5(F)_7(F)_9(F)_11(F)_13(F)_15(F)_17(F)_19(F)_21(E)_22_END	TC69
16	1_3_H1(T)_H2(F)_H7_H1(T)_H2(F)_H7_H1(F)_H8_5(F)_7(F)_9(F)_11(F)_13(F)_15(F)_17(F)_19(F)_21_23(E)_24_26_27_END	N/A

DESCRIPTION	FIELD
Basic path	All
Path considering that the input is not found in the file	N/A
Path considering that the JSON file being read is not JSON formatted	N/A
Path considering a duplicate PROJECT_ID field is found	PROJECT_ID
Path considering a duplicate FILENAME field is found	FILENAME
Path from JSON structure being invalid and PROJECT_ID missing after processing the pairs	PROJECT_ID
Path from an empty JSON file	FIELDS
Path from a missing PROJECT_ID	PROJECT_ID
Path from an invalid PROJECT_ID value	PROJECT_ID
Path from a missing FILENAME	FILENAME
Path from an invalid extension due to missing dot "."	EXTENSION
Path from a file extension not being one of the allowed values	EXTENSION

Path from an extension duplicate	EXTENSION
Path from an invalid NAME value	NAME
Path when there is an error generating the file signature	N/A
Path when reading all_documents.json fails because the file doesn't exist yet	N/A

EXPECTED RESULT	OBSERVATIONS
	F stands for False; T stands for True with condition nodes.
	This is the happy path that executes if a valid JSON file is inputted.
SHA-256 file signature; document data stored in all_documents.json output file	The helper method with a loop to check for duplicate fields is iterated through 2 times (once per field), on the third iteration, the loop ends and returns to the main path.
Input file not found.	(E) represents an exception from a "try" node.
The file is not JSON formatted.	
JSON does not have the expected structure: duplicate field PROJECT_ID	
JSON does not have the expected structure: duplicate field FILENAME	
JSON does not have the expected structure: duplicate field PROJECT_ID	
JSON does not have the expected structure: missing FIELDS	0 iterations in helper method loop.
JSON does not have the expected structure. Invalid PROJECT_ID label	
JSON data has no valid values: invalid PROJECT_ID value	
JSON does not have the expected structure. Invalid FILENAME label	
JSON data has no valid values: invalid EXTENSION	No "." in file name value autamatically means the extension with be invalid (must be .pdf, .docx, or .xlsx)
JSON data has no valid values: invalid EXTENSION	Must be .pdf, .docx, or .xlsx

JSON data has no valid values: invalid EXTENSION	
JSON data has no valid values: invalid NAME	Filename NAME value is invalid and does not satisfy the required alphanumeric format/length
Internal processing error when getting the file_signature.	How to handle an internal processing error when generating the file signature.
SHA-256 file signature; document data stored in all_documents.json output file	FileNotFound error is caught as all_documents.json does not exist, so it is created